

Recitation 03

Decision Structure

Exercise 1 | Boolean Expression

1) Evaluate the following expression

False or True and False	
True and not (9 < 5)	
24 / 12 >= 2 and 'hello'!= 'goodbye'	
55 != 'hello'	
'2'== 2	
'hello'> 35	

2) Assume that `x = 10`, `y = 20`, and `z = 30`. Evaluate the following boolean expressions. If the expression is always `True` or `False` regardless of the values of `x`, `y` and `z`, then add *Always* to your answer. For example:

`x == 10` `True`

`x == x` `Always True`

<code>(x == 10) and (y > 10)</code>	
<code>x < 10 and x > 10</code>	
<code>(x < 10) or (x > 10)</code>	
<code>not (x < (y + z)) or not ((x + 10) <= 20)</code>	
<code>not (x == y) and (x != y) and (x < y or y < x)</code>	

Exercises 2 - String indexing & slicing

```
s = 'Hello World\n'
```

s[-1]		s[5:6]	
s[-2]		s[6:5]	
len(s)		s[-6:-1]	
s[-len(s)]		s[:5:2]	
s[-13]		s[5::-1]	
s[:5]		s[-2:-7:-1]	
s[6:500]		s[:7] + s[7:]	
s[7:]			

Exercise 3 | Areas of Rectangles

The area of a rectangle is the rectangle's length times its width. Write a program that asks for the length and width of two rectangles. The program should tell the user which rectangle has the greater area, or if the areas are the same.

Exercise 4 | Age Classifier

Write a program that asks the user to enter a person's age. The program should display a message indicating whether the person is an infant, a child, a teenager, or an adult.

Following are the guidelines:

- If the person is 1 year old or less, he or she is an infant.
- If the person is older than 1 year, but younger than 13 years, he or she is a child.
- If the person is at least 13 years old, but less than 20 years old, he or she is a teenager.
- If the person is at least 20 years old, he or she is an adult.

Exercise 5 | Roman Numerals

Write a program that prompts the user to enter a number within the range of 1 through 10. The program should display the Roman numeral version of that number. If the number is outside the range of 1 through 10, the program should display an error message.

The following table shows the Roman numerals for the numbers 1 through 10:

Roman number	I	II	III	IV	V	VI	VII	VIII	IX	X
Value	1	2	3	4	5	6	7	8	9	10

Exercise 6 | Color Mixer

The colors red, blue, and yellow are known as the primary colors because they cannot be made by mixing other colors. When you mix two primary colors, you get a secondary color, as shown here:

- When you mix red and blue, you get purple.
- When you mix red and yellow, you get orange.
- When you mix blue and yellow, you get green.

Design a program that prompts the user to enter the names of two primary colors to mix. If the user enters anything other than `red`, `blue` or `yellow` the program should display an error message. Otherwise, the program should display the name of the secondary color that results.

Exercise 7 | Mass and Weight

Scientists measure an object's mass in kilograms and its weight in newtons. If you know the amount of mass of an object in kilograms, you can calculate its weight in newtons with the following formula:

$$Weight = mass \times 9.8$$

Write a program that asks the user to enter an object's mass, then calculates its weight. If the object weighs more than 500 newtons, display a message indicating that it is too heavy. If the object weighs less than 100 newtons, display a message indicating that it is too light.

Exercise 8 | Roulette Wheel Colors

On a roulette wheel, the pockets are numbered from 0 to 36. The colors of the pockets are as follows:

- Pocket 0 is green.
- For pockets 1 through 10, the odd-numbered pockets are red and the even-numbered pockets are black.
- For pockets 11 through 18, the odd-numbered pockets are black and the even-numbered pockets are red.
- For pockets 19 through 28, the odd-numbered pockets are red and the even-numbered pockets are black.
- For pockets 29 through 36, the odd-numbered pockets are black and the even-numbered pockets are red.

Write a program that asks the user to enter a pocket number and displays whether the pocket is green, red, or black. The program should display an error message if the user enters a number that is outside the range of 0 through 36.

Exercise 9 | Software Sales

A software company sells a package that retails for \$99. Quantity discounts are given according to the following table:

Quantity	Discount
10-19	10%
20-49	20%
50-99	30%
100 or more	40%

Exercise 10 | Time calculator

Write a program that asks the user to enter a number of seconds and works as follows:

- There are 60 seconds in a minute.
If the number of seconds entered by the user is greater than or equal to 60, the program should display the number of minutes in that many seconds.
- There are 3,600 seconds in an hour.
If the number of seconds entered by the user is greater than or equal to 3,600, the program should display the number of hours in that many seconds.
- There are 86,400 seconds in a day.
If the number of seconds entered by the user is greater than or equal to 86,400, the program should display the number of days in that many seconds.